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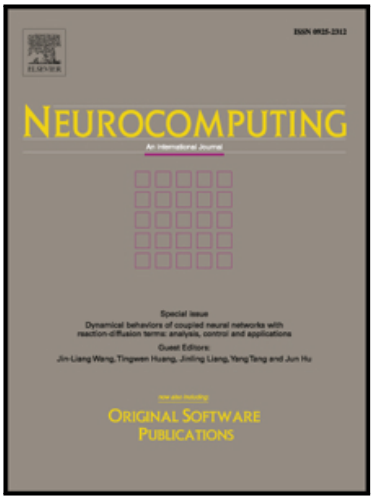

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Disturbance observer based adaptive neural control of uncertain MIMO nonlinear systems with unmodeled dynamics

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Abstract

This paper investigates the disturbance observer-based adaptive neural tracking control of a class of multiple-input multiple-output (MIMO) systems in the presence of unmodeled dynamics, system uncertainties, time varying disturbance and input dead-zone. An adaptive neural control method combined with backstepping technique and the radial basis function neural networks (RBFNNs) is proposed for the systems under consideration. In recursive backstepping designs, neural network (NN) is employed for uncertainty approximation. The disturbance observer is developed to provide efficient learning of the compounded disturbance which includes the effect of time varying disturbance, neural network approximation error. It is shown that by using Lyapunov methods, the developed control scheme can ensure semi-globally uniformly ultimately bounded (SGUUB) of all signals within the closed-loop systems. Simulation results are presented to illustrate the validity of the approach. This paper is novel at least in the two aspects: (1) disturbance observer based tracking control method is developed for MIMO nonlinear systems with unmodeled dynamics and (2) the strong coupled terms is considered in this paper where the interconnections are functions of all states, which is a more general form than existing related results.

Keywords: Adaptive backstepping control, Disturbance observer, Unmodeled dynamics, Input dead-zone, Radial basis function neural networks(RBFNNs)

1. Introduction

The problem of nonlinear is commonly resides in almost all real systems since many systems cannot be simplified into a linear model. Therefore, tremendous attention has paid on the control design of nonlinear systems during the past few decades. A variety of remarkable control approaches have been presented in the literature, including adaptive control[1–3], fuzzy control [4–6], fault tolerant control [7, 8] and intelligent control [9–11]. Among them, the adaptive backstepping technique as an effective control approach has undergone rapid development and played an important role in the control of nonlinear systems [1–3]. For a class of complex nonlinear systems with unknown functions, many control schemes were explored by combining adaptive control and functional approximators. It is konwn that neural networks and fuzzy-logic systems are proved to be effective approaches for controlling the highly uncertain nonlinear systems [12–23]. In general, neural networks or fuzzy systems are employed to estimate uncertain continuous nonlinear functions, to list a few, in [24, 25], the adaptive control based neural networks was developed for nonlinear systems with unknown virtual control coefficient functions. In [26], an effective neural network control is developed for Bimanual Robots systems with unknown uncertainties. In [20, 27], the research is extended to multiinput multioutput (MIMO) nonlinear systems. Alternatively, many researchers have been actively working on the fuzzy systems-based control design of uncertain nonlinear systems containing unknown functions [4, 28–31].

Input dead-zone problem exists in many real systems, their existences frequently deteriorates control performances and even result in the instability of the system. Therefore, the consideration on the control of nonlinear systems affected by input dead-zone has accepted increasing attention, and a variety of effective control methods have been proposed in [32–36]. In [33], the robust adaptive control is presented for a class of uncertain pure-feedback nonlinear systems with actuator dead-zones by using fuzzy methods. A class of nonlinear uncertain systems with dead-zone has been solved in the work [34], and the basic idea was to treat dead-zone as disturbance in a same way. Moreover, the author in [35] solved the chattering problems by proposing a smooth adaptive dead-zone inverse algorithm and a different adaptive compensation scheme for unknown dead-zone is derived in [36]. However, it is observed that the unmodeled dynamics is not seriously considered based on the aforementioned results.

On the other hand, the unmodeled dynamics and unknown external disturbance often exist in actual engineering, since the practical systems are almost impossible to describe precise mathematical models. Therefore, the consideration on the control of nonlinear systems with unmodeled dynamics has gained

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increasing attention, and a great number of valuable control algorithms have been presented in [22, 23, 37–41]. In [22], an adaptive neural output-feedback control method was presented for a class of nonlower triangular nonlinear systems with unmodeled dynamics. In [23], an adaptive neural tracking control problem was considered for a class of nonlower triangular nonlinear systems with unmodeled dynamics. By introducing modified dynamic signal, the authors in [39] proposed the robust adaptive control to deal with the unmodeled dynamics. In [42], an adaptive decentralized control method was proposed for strong interconnected nonlinear systems with unmodeled dynamics using backstepping technique. Recently, the corresponding researches have been extended to interconnected nonlinear systems with unmodeled dynamics. The work in [41, 43] proposed a fuzzy decentralized control approach based observer for interconnected systems with unmodeled dynamics. However, the interconnection terms in [41, 43] are only the functions of each subsystem, cannot be practical. Therefore, it is a meaningful research topic to design an adaptive backstepping control approach for nonlinear systems with unmodeled dynamics where the interconnection terms are functions of all states. Meanwhile, some significant achievements focusing on problems of external disturbances have also been reported in the literature [44–49]. In [44], the disturbance observer based sliding model control is designed for vehicular radar system. In [45], disturbance observer based adaptive output feedback control is proposed for strict-feedback systems with time-varying disturbance. In [46], the the optimal control problem is investigated based the disturbance-observer for a class of delta-domain networked control systems. In [48], the dynamic surface design with disturbance observer is provided for the transport aircraft. In [49], the composite control problem is considered based the disturbance-observer for a class of delta domain linear quadratic game with both matched and unmatched disturbances. As a matter of fact, many control issues of MIMO uncertain nonlinear systems are still a challenging task, and enhance the design difficulty due to the existence of input dead-zone, unmodeled dynamics, system uncertainty, and external unknown disturbances, which needs to be further studied.

With the statements above, in this paper, the disturbance observer-based adaptive neural tracking control is developed for a class of MIMO systems in the presence of unmodeled dynamics, system uncertainties, time varying disturbance and input dead-zone. Compared with the existing results, the main contributions are summarised as follows:

(1) Compared with the existing works on the control of nonlinear systems with unmodeled dynamics which mainly considered the stabilized control, e.g., [50, 51], while in this paper, disturbance observer based tracking control method is developed for MIMO nonlinear systems with unmodeled dynamics, (2) the strong coupled terms are considered in this paper where the interconnections are functions of all states, which is in a more general form than existing related results [41, 43], (3) the coupler design between the RBFNNs and disturbance observer is considered in this paper, which improves the functional approximation capability and disturbance compensation ability by comparing with the existing works [44].

The rest of the paper is organised as follows: The dynamic model of n-order uncertain nonlinear systems with input dead-zone is presented in Section 2. An adaptive RBFNN control strategy combined disturbance observer with backstepping method is presented in Section 3. Simulation results are provided in Section 4. Section 5 presents the conclusions of this paper.

2. Problem formulation and preliminaries

Consider a class of nonlinear MIMO continuous time strict-feedback systems with subsystems subject to unknown dead-zone described by the following differential equations:

$$\begin{cases} \dot{z}_i = q_i(z_i, x_i), \\ \dot{x}_{ij} = g_{ij}(\bar{x}_{ij})x_{ij+1} + f_{ij}(X) + \Delta_{ij}(x_i, z_i, t) + d_{ij}(t), \\ \dot{x}_{in_i} = g_{in_i}u_i(v_i) + f_{in_i}(X) + \Delta_{in_i}(x_i, z_i, t) + d_{in_i}(t), \\ y_i = x_{i1}, \end{cases} \tag{1}$$

where $i = 1, 2, \ldots, N$; $j = 1, \ldots, n_i - 1$; $x_i = \bar{x}_{in_i} = [x_{i1}, \ldots, x_{in_i}]^T \in R^{n_i}$ and $X = [\bar{x}_1^T, \ldots, \bar{x}_N^T]^T$ stand for the state of the ith subsystem and overall system, respectively, $\bar{x}_{ij} = [x_{i1}, \ldots, x_{ij}]^T \in R^j$; $y_i \in R$ stands for the output in the ith subsystem; z_i is unmodeled dynamics; $g_{ij}(\cdot)$ and $f_{ij}(\cdot)$ represent the unknown nonlinear functions; $\Delta_{ij}(\cdot)$ denotes the system dynamic uncertainties; and $\Delta_{ij}(\cdot)$ and $q_i(\cdot)$ are assumed to be uncertain Lipschitz continuous functions; $d_{ij}(t)$ stands for the unknown external disturbance; $u_i \in R$ is the control input in the i-th subsystem, and the output of the unknown dead-zone which is described by

$$u_i = G(v_i(t)) = \begin{cases} g_r(v_i), & \text{if } v_i(t) \geq d_r \\ 0, & \text{if } d_l \leq v_i(t) \leq d_r \\ g_l(v_i), & \text{if } v_i(t) \leq d_l \end{cases} \tag{2}$$

where v_i is the input of dead zone, d_r and d_l are the unknown constants, and $d_r > 0, d_l < 0$.

For uncertain MIMO nonlinear system (1) the objective of this paper is to design an adaptive tracking controller such that the output y tracks a target signal y_r within a bounded error and all the signals are bounded under the existence of system uncertainties, input dead zone and unknown external disturbances.

According to the analysis in [35], the solution of the system (2) can be transformed into

$$u_i = G(v_i(t)) = c_i v_i(t) + \iota_i(v_i) \tag{3}$$

where c_i is the positive constant, $\iota_i(v_i)$ is bounded and satisfying $|\iota_i(v_i)| \leq \bar{\iota}_i$, $\bar{\iota}_i$ is unknown positive constant.

Assumption 1. [45] The target signal and its time derivatives up to the nth order are continuous and uniformly bounded. There exists a positive constant Y_{i0} satisfy $|y_{ri}(t)| \leq Y_{i0}$ for all $t \geq 0$, $i = 1, \ldots, N$.

Assumption 2. [52] The unknown external disturbance $D_i(t) = [d_{i1}(t), \ldots, d_{in_i}(t)]^T$ is bounded and satisfies $\left|d_{ij}(t)\right| \leq \rho_{ij}$, where $\rho_{ij} > 0$ is unknown, $i = 1, 2, \ldots, N$, $j = 1, 2, \ldots, n_i$.

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Assumption 3. [4] The functions $g_{ij}(\cdot)$ are known, and there exists a positive constant g_0 satisfying

$$0 < g_0 \leq \left|g_{ij}(\cdot)\right| < \infty \tag{4}$$

Assumption 4. [41, 53] For the dynamic uncertainties Δ_{ij}, there are unknown nonnegative smooth functions $\varphi_{ij1}(\cdot)$ and $\varphi_{ij2}(\cdot)$ which satisfies

$$\left|\Delta_{ij}\right| \leq \varphi_{ij1}(\left|(\bar{x}_{ij})\right|) + \varphi_{ij2}(|z_i|) \tag{5}$$

Remark 1. Assumption 4 relaxes the restriction in the existing results [41, 53], in which $\varphi_{ij1}(\cdot)$ and $\varphi_{ij2}(\cdot)$ are not required to be known in this paper.

Definition 1. [53] If a continuous function $\zeta : [0, a) \to [0, \infty)$ is strictly increasing and $\zeta(0) = 0$, then the function ζ is called K function. If $a = \infty$ and $r \to \infty, \zeta(r) \to \infty$, then ζ is called K_∞ function.

Assumption 5. [41, 51] The unmodeled dynamics in (1) is exponentially input-to-state practically stable (exp-ISpS); that is, for the system $\dot{z}_i = q(z_i, x_i)$, there exists an exp-ISpS Lyapunov function $V_i(z_i)$ such that

$$\begin{aligned} &\alpha_{i1}(|z_i|) \leq V_i(z_i) \leq \alpha_{i2}(|z_i|) \\ &\frac{\partial V_i(z_i)}{\partial z_i} q_i(z_i, x_i) \leq -c_i V_i(z_i) + \mu_i(|x_i|) + d_i \end{aligned} \tag{6}$$

where α_{i1}, α_{i2}, and μ_i are of class K_∞-functions and c_i and d_i are known positive constants.

Lemma 1. [38] If V_i is an exp-ISpS Lyapunov function for a control system, then, for any constants $\bar{c}_i$ in $(0, c_{i0})$, any initial condition $x_{i0} = x_{i0}(0)$, and any function $\bar{\mu}(x_{i1}) \geq \mu(|x_{i1}|)$, there exists a finite time $T_{i0} = T_{i0}(\bar{c}_i, r_{i0}, z_{i0})$, a nonnegative function $D_i(t)$ defined for all $t \geq 0$, and a signal described by

$$\dot{r}_i = -\bar{c}_i r_i + \bar{\mu}_i(|x_{i1}|) + d_i, r_i(0) = r_{i0}, \tag{7}$$

such that $D_i(t) = 0$ for all $t \geq T_{i0}$,

$$V_i(z_i(t)) \leq r_i(t) + D_i(t) \tag{8}$$

for all $t \geq 0$. Takes $\bar{\mu}_i(x_{i1}) = x_{i1}^2 \mu_{i0}(x_{i1}^2)$, where $\bar{\mu}_{i0}(\cdot)$ is a nonnegative smooth function, and μ_{i0} is a nonnegative smooth function. The dynamical r_i can be rewritten as

$$\dot{r}_i = -\bar{c}_i r_i + x_{i1}^2 \mu_{i0}(x_{i1}^2) + d_i, r_i(0) = r_{i0} \tag{9}$$

Lemma 2. [13] The following inequality holds for any $\epsilon > 0$ and for any $\eta \in R$

$$0 \leq |\eta| - \eta \tanh\left(\frac{\eta}{\epsilon}\right) \leq \kappa\epsilon \tag{10}$$

where κ is a constant satisfying $\kappa = e^{-(\kappa+1)}$, i.e., $\kappa \approx 0.2785$.

Lemma 3. [54] Consider the set $\Omega_{z_{i1}}$ defined by $\Omega_{z_{i1}} := \{z_{i1} \| z_{i1}| < 0.8814\tau_i\}$. Then, for any $z_{i1} \notin \Omega_{z_{i1}}$, the inequality $\left(1 - 2\tanh^2(z_{i1}/\tau_i)\right) \leq 0$ is satisfied.

Lemma 4. [14] The radial basis function has the following form :

$$h_{ijq}(Z_i) = \exp\left[\frac{-(Z_i - \mu_{ijq})^T (Z_i - \mu_{ijq})}{\eta_i^2}\right], q = 1, \ldots . l_{ij} \tag{11}$$

where η_i is the width of the Gaussian function, and $\mu_{ijq} = [\mu_{ij1}, \mu_{ij2}, \ldots, \mu_{ijl_{ij}}]^T$ is the center vector. A continuous nonlinear function $f_{ij}(Z_i)$ can be approximated by an ideal RBF neural network on a compact set $Z_i \in \Omega_{Z_i} \subset R^q$ with the neural node number $l_{ij} > 1$. For any $\xi_{ij} > 0$, the following inequality holds:

$$\sup_{Z_i \in \Omega} \left|f_{ij}(Z_i) - W_{ij}^T H_{ij}(Z_i)\right| \leq \xi_{ij} \tag{12}$$

where $H_{ij}(Z_i) = [h_{ij1}(Z_i), \ldots, h_{ijlij}(Z_i)]^T$ is the basis function vector, $W_{ij} = [w_{ij1}, \ldots, w_{ijlij}]^T$ is the ideal constant weight vector. In this paper, we directly estimate the norm of W_{ij} instead of themselves, so the θ_{ij} can be expressed as $\left\|W_{ij}\right\|^2 = \theta_{ij}$, $\hat{\theta}_{ij}$ is the estimate of θ_{ij}, $\tilde{\theta}_{ij} = \theta_{ij} - \hat{\theta}_{ij}$ is the estimate error.

Remark 2. In general, it should be emphasized that the selection of the centers and widths of RBF has a great influence on the performance of the designed controller. According to [55, 56], (1) increasing the NN node number l will help to reduce the inherent NN approximation error ξ; (2) the centers μ_{ij} evenly cover the NN approximation domain; (3) the widths η_i of RBFs are calculated by using the formula $\eta = \sqrt{2}d_{max}/\sqrt{l}$ where d_{max} is the maximal Euclid distance of the approximation domain. More details on selecting these parameters can be found in [55, 56].

3. Adaptive control design via disturbance observer

An adaptive RBFNN control strategy combined disturbance observer with backstepping method is presented in this section. Firstly, the following change of coordinate is defined:

Step 1: Firstly, the following change of coordinate is defined for the first subsystem:

$$z_{i,1} = x_{i1} - y_{ri} \tag{13}$$

$$z_{i,2} = x_{i2} - \alpha_{i1} \tag{14}$$

Under the coordinate transformation (13) and (14), the time derivative of z_{i1} is given by

$$\begin{aligned} \dot{z}_{i1} =& \dot{x}_{i1} - \dot{y}_{ri} \\ =& g_{i1}(\alpha_{i1} + z_{i2}) + f_{i1} + \Delta_{i1}(x_i, z_i, t) + d_{i1} - \dot{y}_{ri} \end{aligned} \tag{15}$$

Consider a Lyapunov function candidate as

$$V_{i1} = \frac{1}{2}z_{i1}^2 + \frac{r_i}{\lambda_{i0}} + \frac{1}{2r_{i1}}\tilde{\theta}_{i1}^2 + \frac{1}{2}\tilde{\mathcal{D}}_{i1}^2 \tag{16}$$

where $r_{i1} > 0$ is a design parameter, $\tilde{\mathcal{D}}_{i1} = \mathcal{D}_{i1} - \hat{\mathcal{D}}_{i1}$ is the estimate error of disturbance. By Assumption 4 and Lemma 1,

the derivative of V_{i1} is given by

$$\begin{aligned}\dot{V}_{i1} &= z_{i1}\dot{z}_{i1} + \frac{1}{\lambda_{i0}}\dot{r}_i - \frac{1}{r_{i1}}\tilde{\theta}_{i1}\dot{\hat{\theta}}_{i1} + \tilde{\mathcal{D}}_{i1}\dot{\tilde{\mathcal{D}}}_{i1}\\
&= z_{i1}\Big(g_{i1}(\alpha_{i1} + z_{i2}) + f_{i1}(X) + \Delta_{i1} + d_{i1} - \dot{y}_{ri}\Big)\\
&\quad + \frac{1}{\lambda_{i0}}\dot{r}_i - \frac{1}{r_{i1}}\tilde{\theta}_{i1}\dot{\hat{\theta}}_{i1} + \tilde{\mathcal{D}}_{i1}\dot{\tilde{\mathcal{D}}}_{i1}\\
&\le z_{i1}\Big(g_{i1}\alpha_{i1} + g_{i1}z_{i2} + f_{i1}(X) + d_{i1} - \dot{y}_{ri} + \frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}z_{i1}}\Big)\\
&\quad + |z_{i1}||\Delta_{i1}| + \frac{d_{i0}}{\lambda_{i0}} - \frac{\bar{c}_i}{\lambda_{i0}}r_i - \frac{1}{r_{i1}}\tilde{\theta}_{i1}\dot{\hat{\theta}}_{i1} + \tilde{\mathcal{D}}_{i1}\dot{\tilde{\mathcal{D}}}_{i1}\\
&\le z_{i1}\Big(g_{i1}\alpha_{i1} + g_{i1}z_{i2} + f_{i1}(X) + d_{i1} - \dot{y}_{ri} + \frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}z_{i1}}\Big)\\
&\quad + |z_{i1}|\varphi_{i11}(|x_{i1}|) + |z_{i1}|\varphi_{i12}(|z_i|) + \frac{d_{i0}}{\lambda_{i0}} - \frac{\bar{c}_i}{\lambda_{i0}}r_i\\
&\quad - \frac{1}{r_{i1}}\tilde{\theta}_{i1}\dot{\hat{\theta}}_{i1} + \tilde{\mathcal{D}}_{i1}\dot{\tilde{\mathcal{D}}}_{i1} \qquad (17)\end{aligned}$$

By using Lemma 2, the following inequality holds:

$$\begin{aligned}|z_{i1}|\varphi_{i11}(|x_{i1}|) &\le z_{i1}\varphi_{i11}(|x_{i1}|)\tanh\left(\frac{z_{i1}\varphi_{i11}(|x_{i1}|)}{\epsilon_{i11}}\right) + \epsilon'_{i11}\\
&\le z_{i1}\hat{\varphi}_{i11}(x_{i1}) + \epsilon'_{i11} \qquad (18)\end{aligned}$$

where $\hat{\varphi}_{i11}(x_{i1}) = \varphi_{i11}(|x_{i1}|)\tanh\left(\frac{z_{i1}\varphi_{i11}(|x_{i1}|)}{\epsilon_{i11}}\right)$ is a smooth function.
Similar to the derivations in [38], we have

$$\begin{aligned}|z_{i1}|\varphi_{i12}(|z_i|) &\le |z_{i1}|\varphi_{i12}\circ\alpha_1^{-1}(2r_i) + \frac{1}{4}z_{i1}^2 + h_{i1}(t)\\
&\le |z_{i1}|\bar{\varphi}_{i12}(r_i) + \frac{1}{4}z_{i1}^2 + h_{i1}(t)\\
&\le z_{i1}\bar{\varphi}_{i12}(r_i)\tanh\left(\frac{z_{i1}\bar{\varphi}_{12}(r_i)}{\epsilon_{i12}}\right) + \epsilon'_{i12} + \frac{1}{4}z_{i1}^2 + h_{i1}(t)\\
&\le z_{i1}\hat{\varphi}_{i12}(r_i) + \epsilon'_{i12} + \frac{1}{4}z_{i1}^2 + h_{i1}(t) \qquad (19)\end{aligned}$$

where $\varphi_{i12}\circ\alpha_{i1}^{-1}(2r_i) = \varphi_{i12}[\alpha_{i1}^{-1}(2r_i)]$, $\bar{\varphi}_{i12}(r_i) = \varphi_{i12}\circ\alpha_{i1}^{-1}(2r_i)$, $\hat{\varphi}_{i12}(r_i) = \bar{\varphi}_{i12}(r_i)\tanh\left(\frac{z_{i1}\bar{\varphi}_{i12}(r_i)}{\epsilon_{i12}}\right)$, $h_{i1}(t) = (\varphi_{i12}\circ\alpha_{i1}^{-1}(2D_i(t)))^2$, $\varphi_{i12}\circ\alpha_{i1}^{-1}(2D_i(t)) = \varphi_{i12}[\alpha_{i1}^{-1}(2D_i(t))]$ are smooth function. Since the term $\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}z_{i1}}$ in (17) is discontinuous at $z_{i1} = 0$, we introduce a hyperbolic tangent function $\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)$, and substituting (18) and (19) into (17) results in

$$\begin{aligned}\dot{V}_1 &\le z_{i1}\left(g_{i1}\alpha_{i1} + g_{i1}z_{i2} + \hat{f}_{i1}(Z_{i1})\right) + z_{i1}d_{i1} + \frac{d_{i0}}{\lambda_{i0}} - \frac{\bar{c}_i}{\lambda_{i0}}r_i\\
&\quad + \left(1 - 2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}} - \frac{1}{r_{i1}}\tilde{\theta}_{i1}\dot{\hat{\theta}}_{i1}\\
&\quad + \tilde{\mathcal{D}}_{i1}\dot{\tilde{\mathcal{D}}}_{i1} + \sum_{k=1}^{2}\epsilon'_{i1k} + h_{i1}(t) \qquad (20)\end{aligned}$$

where τ_i is a given positive constant and unknown nonlinear function $\hat{f}_{i1}(Z_{i1})$ defined as

$$\begin{aligned}\hat{f}_{i1}(Z_{i1}) &= f_{i1}(X) - \dot{y}_{ri} + \hat{\varphi}_{i11}(x_{i1}) + \hat{\varphi}_{i12}(r_i) + \frac{1}{4}z_{i1}\\
&\quad + \frac{2}{z_{i1}}\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\frac{x_{i1}^2\mu_{i0}(x_{i1}^2)}{\lambda_{i0}},\\
Z_{i1} &= [x_{i1}, r]^T \in \Omega_{Z_{i1}} \subset R^2. \qquad (21)\end{aligned}$$

Additionally, the remaining term $\left(1 - 2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}}$ in (20) will be dealt with at the end of this section. By employing an RBFNN $W_{i1}^T H_{i1}(Z_{i1})$ to approximate $\hat{f}_{i1}(Z_{i1})$, the following equation can be obtained

$$\hat{f}_{i1}(Z_{i1}) = W_{i1}^T H_{i1}(Z_{i1}) + \xi_{i1} \qquad (22)$$

where ξ_{i1} is the approximation error. Since there exists RBFNN approximation errors to be suppressed, and external time-varying disturbance to be compensated, we need to design a disturbance observer. Then, define the disturbance observer variable as

$$\mathcal{D}_{i1} = d_{i1} + \xi_{i1} \qquad (23)$$

Considering Assumption 2 and the RBFNN approximation theory, we have $\|\dot{\mathcal{D}}_{i1}\| \le \rho_{i1}$, where $\rho_{i1} > 0$ is an unknown constant. Define the auxiliary variable as

$$\Xi_{i1} = \mathcal{D}_{i1} - c_{i1}z_{i1} \qquad (24)$$

By using (24), we can obtain the estimate of disturbance $\mathcal{D}_{i1}$ as follows:

$$\hat{\mathcal{D}}_{i1} = \hat{\Xi}_{i1} + c_{i1}z_{i1} \qquad (25)$$

The following equation is proposed based on (24) to estimate Ξ_{i1}:

$$\dot{\hat{\Xi}}_{i1} = -c_{i1}\left(\hat{\theta}_{i1}H_{i1}^T H_{i1} + z_{i2} + \alpha_{i1} + \hat{\mathcal{D}}_{i1}\right) \qquad (26)$$

Calculating the derivative of the estimate error of disturbance $\tilde{\mathcal{D}}_{i1}$ as follows:

$$\dot{\tilde{\mathcal{D}}}_{i1} = \dot{\mathcal{D}}_{i1} - c_{i1}\left(\tilde{\mathcal{D}}_{i1} - \tilde{\theta}_{i1}H_{i1}^T H_{i1}\right) \qquad (27)$$

Then, the virtual control input and adaption laws shall be designed as

$$\alpha_{i1} = g_{i1}^{-1}\left(-k_{i1}z_{i1} - \frac{z_{i1}}{2\gamma_{i1}^2}\hat{\theta}_{i1}H_{i1}^T H_{i1} - \hat{\mathcal{D}}_{i1} - \frac{z_{i1}}{2} + \dot{y}_{ri}\right) \qquad (28)$$

$$\dot{\hat{\theta}}_{i1} = r_{i1}\left(\frac{1}{2\gamma_{i1}^2}H_{i1}^T H_{i1}z_{i1}^2 - \sigma_{i1}\hat{\theta}_{i1}\right) \qquad (29)$$

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where σ_{i1} is the positive design parameters. Substituting (21)-(29) in to (20) results in

$$\begin{aligned}\dot{V}_{i1} \leq & -k_{i1}z_{i1}^2 + \underbrace{\sigma_{i1}\tilde{\theta}_{i1}\hat{\theta}_{i1}}_{\mathrm{I}} + g_{i1}z_{i1}z_{i2} \\ & + \underbrace{z_{i1}\tilde{\mathcal{D}}_{i1} + \tilde{\mathcal{D}}_{i1}\dot{\mathcal{D}}_{i1} + c_{i1}\tilde{\mathcal{D}}_{i1}\tilde{\theta}_{i1}H_{i1}^T H_{i1}}_{\mathrm{II}} - c_{i1}\tilde{\mathcal{D}}_{i1}^2 \\ & + \left(1 - 2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}} + \frac{d_{i0}}{\lambda_{i0}} - \frac{\bar{c}_i}{\lambda_{i0}}r_i \\ & + \sum_{k=1}^{2}\epsilon'_{i1k} + h_{i1}(t) + \frac{\gamma_{i1}^2}{2} - \frac{z_{i1}^2}{2}\end{aligned} \tag{30}$$

The following inequalities hold

$$\text{(I)}: \sigma_{i1}\tilde{\theta}_{i1}\hat{\theta}_{i1} \leq -\frac{\sigma_{i1}\tilde{\theta}_{i1}^2}{2} + \frac{\sigma_{i1}\theta_{i1}^2}{2} \tag{31}$$

$$\begin{aligned}\text{(II)}: & z_{i1}\tilde{\mathcal{D}}_{i1} \leq \frac{z_{i1}^2}{2} + \frac{\tilde{\mathcal{D}}_{i1}^2}{2} \\ & \tilde{\mathcal{D}}_{i1}\dot{\mathcal{D}}_{i1} \leq \frac{\tilde{\mathcal{D}}_{i1}^2}{2} + \frac{\rho_{i1}^2}{2} \\ & c_{i1}\tilde{\mathcal{D}}_{i1}\tilde{\theta}_{i1}H_{i1}^T H_{i1} \leq \frac{\tilde{\mathcal{D}}_{i1}^2}{2} + \frac{\mu_{i1}^2 c_{i1}^2\tilde{\theta}_{i1}^2}{2}\end{aligned} \tag{32}$$

Based on (31) and (32), Eq. (30) satisfies the following inequality

$$\begin{aligned}\dot{V}_{i1} \leq & -k_{i1}z_{i1}^2 - \frac{\sigma_{i1} - \mu_{i1}^2 c_{i1}^2}{2}\tilde{\theta}_{i1}^2 + g_{i1}z_{i1}z_{i2} - (c_{i1} - 1.5)\tilde{\mathcal{D}}_{i1}^2 \\ & + \left(1 - 2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}} + \frac{d_{i0}}{\lambda_{i0}} - \frac{\bar{c}_i}{\lambda_{i0}}r_i \\ & + \sum_{k=1}^{2}\epsilon'_{i1k} + h_{i1}(t) + \frac{\rho_{i1}^2}{2} + \frac{\gamma_{i1}^2}{2} + \frac{\sigma_{i1}\theta_{i1}^2}{2}\end{aligned} \tag{33}$$

Step j $(j = 2, \ldots, n_i - 1)$: The following change of coordinate is defined:

$$z_{ij} = x_{ij} - \alpha_{i(j-1)}, j = 2, \ldots, n_i - 1 \tag{34}$$

Consider a Lyapnov function candidate as

$$V_{ij} = \frac{1}{2}z_{ij}^2 + \frac{1}{2r_{ij}}\tilde{\theta}_{ij}^2 + \frac{1}{2}\tilde{\mathcal{D}}_{ij}^2 \tag{35}$$

Differentiating (35) results in

$$\begin{aligned}\dot{V}_{ij} = & z_{ij}\Big(g_{ij}(\alpha_{ij} + z_{i(j+1)}) + f_{ij}(X) + \Delta_{ij} + d_{ij} - \dot{\alpha}_{i(j-1)}\Big) \\ & - \frac{1}{r_{ij}}\tilde{\theta}_{ij}\dot{\hat{\theta}}_{ij} + \tilde{\mathcal{D}}_{ij}\dot{\tilde{\mathcal{D}}}_{ij}\end{aligned} \tag{36}$$

where

$$\begin{aligned}\dot{\alpha}_{i(j-1)} = & \sum_{k=1}^{j-1}\left\{\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\Big(g_{ik}x_{i,(k+1)} + f_{ik}(X) + \Delta_{ik} + d_{ik}\Big)\right\} \\ & + \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial\hat{\theta}_{ik}}\dot{\hat{\theta}}_{ik} + \frac{\partial\alpha_{i(j-1)}}{\partial r_i}\dot{r}_i\end{aligned} \tag{37}$$

Then, $\dot{V}_{ij}$ can be rewritten as

$$\begin{aligned}\dot{V}_{ij} \leq & z_{ij}\Big(g_{ij}(\alpha_{ij} + z_{i(j+1)}) - \sum_{k=1}^{j-1}\frac{\partial\alpha_{i,(j-1)}}{\partial x_{ik}}g_{ik}x_{i,(k+1)} + f_{ij}(X) \\ & - \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}f_{ik}(X) - \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}d_{ik} \\ & + \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial\hat{\theta}_{ik}}\dot{\hat{\theta}}_{ik} + \frac{\partial\alpha_{i(j-1)}}{\partial r_i}\dot{r}_i\Big) + \left|z_{ij}\bar{\Delta}_{ij}\right| + z_{ij}d_{ij} \\ & - \frac{1}{r_{ij}}\tilde{\theta}_{ij}\dot{\hat{\theta}}_{ij} + \tilde{\mathcal{D}}_{ij}\dot{\tilde{\mathcal{D}}}_{ij}\end{aligned} \tag{38}$$

where $\bar{\Delta}_{ij} = \Delta_{ij} - \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\Delta_{ik}$.

By using the similar estimation methods in Step 1, one has

$$\begin{aligned}\left|z_{ij}\bar{\Delta}_{ij}\right| \leq & \left|z_{ij}\right|\left(\left|\Delta_{ij}\right| + \sum_{k=1}^{j-1}\left|\frac{\partial\alpha_{i,(j-1)}}{\partial x_{ik}}\right|\left|\Delta_{ik}\right|\right) \\ \leq & \left|z_{ij}\right|\left(\varphi_{ij1}(\bar{x}_{ij}) + \sum_{k=1}^{j-1}\left|\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\right|\varphi_{ik1}(\bar{x}_{ij})\right) \\ & + \left|z_{ij}\right|\left(\varphi_{ij2}(|z_i|) + \sum_{k=1}^{j-1}\left|\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\right|\varphi_{ik2}(|z_i|)\right)\end{aligned} \tag{39}$$

Similar to (18) and (19), we can obtain

$$\begin{aligned}& \left|z_{ij}\right|\left(\varphi_{ij1}\left(\bar{x}_{ij}\right) + \sum_{k=1}^{j-1}\left|\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\right|\varphi_{ik1}(\bar{x}_{ij})\right) \\ & \leq z_{ij}\hat{\varphi}_{ij1}\left(\bar{x}_{ij}\right) + \epsilon'_{ij1}\end{aligned} \tag{40}$$

$$\begin{aligned}& \left|z_{ij}\right|\left(\varphi_{ij2}(|z_i|) + \sum_{k=1}^{j-1}\left|\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\right|\varphi_{ik2}(|z_i|)\right) \\ & \leq z_{ij}\hat{\varphi}_{ij2}\left(\bar{x}_{ij}, r_i\right) + \frac{z_{ij}^2}{4}\left[1 + \sum_{k=1}^{j-1}\left(\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\right)^2\right] \\ & + \epsilon'_{ij2} + h_{ij}(t)\end{aligned} \tag{41}$$

where $\hat{\varphi}_{ij1}\left(\bar{x}_{ij}\right) = \left(\varphi_{ij1}\left(\bar{x}_{ij}\right) + \sum_{k=1}^{j-1}\left|\frac{\partial\alpha_{i,(j-1)}}{\partial x_{ik}}\right|\varphi_{ik1}\left(\bar{x}_{ij}\right)\right)\tanh\left(z_{ij}\left(\varphi_{ij1} + \sum_{k=1}^{j-1}\left|\partial\alpha_{i(j-1)}/\partial x_{ik}\right|\varphi_{ij1}\right)/\epsilon_{ij1}\right)$, $\hat{\varphi}_{ij2}\left(\bar{x}_{ij}, r_i\right) = \bar{\varphi}_{ij2}\left(\bar{x}_{ij}, r_i\right)\tanh\left(z_{ij}\bar{\varphi}_{ij2}\left(\bar{x}_{ij}, r_i\right)/\epsilon_{ij2}\right)$, $\bar{\varphi}_{ij2}\left(\bar{x}_{ij}, r_i\right) = \varphi_{ij2} \circ \alpha_{i1}^{-1}(2r_i) + \sum_{k=1}^{j-1}\left|\partial\alpha_{i(j-1)}/\partial x_{ik}\right|\varphi_{ij2} \circ \alpha_{i1}^{-1}(2r_i)$, and $h_{ij}(t) = \sum_{k=1}^{j}(\varphi_{ij2} \circ \alpha_{i1}^{-1}(2D_i(t)))^2$, noting that $h_{ij}(t) \geq 0$ for all $t \geq 0$. Substituting (40) and (41) into (38) yields

$$\begin{aligned}\dot{V}_{ij} \leq & z_{ij}\Big(g_{ij}(\alpha_{ij} + z_{i(j+1)}) + \hat{f}_{ij}(Z_{ij})\Big) + z_{ij}d_{ij} \\ & - \frac{1}{r_{ij}}\tilde{\theta}_{ij}\dot{\hat{\theta}}_{ij} + \tilde{\mathcal{D}}_{ij}\dot{\tilde{\mathcal{D}}}_{ij} + \sum_{k=1}^{2}\epsilon'_{ijk} + h_{ij}(t)\end{aligned} \tag{42}$$

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where

$$\begin{aligned}\hat{f}_{ij}(Z_{ij}) =& f_{ij}(X) - \sum_{k=1}^{j-1}\frac{\partial\alpha_{i,(j-1)}}{\partial x_{ik}}g_{ik}x_{i,(k+1)} - \sum_{k=1}^{j-1}\frac{\partial\alpha_{i,(j-1)}}{\partial x_{ik}}f_{ik}(X)\\ &- \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}d_{ik} + \sum_{k=1}^{j-1}\frac{\partial\alpha_{i(j-1)}}{\partial\hat{\theta}_{ik}}\dot{\hat{\theta}}_{ik} + \frac{\partial\alpha_{i(j-1)}}{\partial r_i}\dot{r}_i\\ &+ \hat{\varphi}_{ij1}\left(\bar{x}_{ij}\right) + \hat{\varphi}_{ij2}\left(\bar{x}_{ij}, r_i\right) + \frac{z_{ij}^2}{4}\left[1 + \sum_{k=1}^{j-1}\left(\frac{\partial\alpha_{i(j-1)}}{\partial x_{ik}}\right)^2\right]\\ Z_{ij} =& [\bar{x}_{ij}, \bar{\hat{\theta}}_{i(j-1)}, r]^T \in \Omega_{Z_{ij}} \subset R^{2j}.\end{aligned} \tag{43}$$

By employing an RBFNN $W_{ij}^T H_{ij}(Z_{ij})$ to approximate $\hat{f}_{ij}(Z_{ij})$, the following equation can be obtained

$$\hat{f}_{ij}(Z_{ij}) = W_{ij}^T H_{ij}(Z_{ij}) + \xi_{ij} \tag{44}$$

Define the following disturbance observer:

$$\begin{cases}\mathcal{D}_{ij} = d_{ij} + \xi_{ij}\\ \hat{\mathcal{D}}_{ij} = \hat{\Xi}_{ij} + c_{ij}z_{ij}\\ \dot{\hat{\Xi}}_{ij} = -c_{ij}\left(\hat{\theta}_{ij}H_{ij}^T H_{ij} + g_{ij}(z_{i(j+1)} + \alpha_{ij}) + \hat{\mathcal{D}}_{ij}\right)\\ \tilde{\mathcal{D}}_{ij} = \dot{\mathcal{D}}_{ij} - c_{ij}\left(\tilde{\mathcal{D}}_{ij} - \tilde{\theta}_{ij}H_{ij}^T H_{ij}\right)\end{cases} \tag{45}$$

where $\tilde{\mathcal{D}}_{ij} = \mathcal{D}_{ij} - \hat{\mathcal{D}}_{ij}$ is the estimate error of disturbance, $\dot{\mathcal{D}}_{ij} \le \rho_{ij}, \rho_{ij} > 0$ is an unknown constant, Ξ_{ij} is the auxiliary variable. Then, the virtual control input and adaption laws shall be designed as

$$\alpha_{ij} = g_{ij}^{-1}\left(-k_{ij}z_{ij} - \frac{z_{ij}}{2\gamma_{ij}^2}\hat{\theta}_{ij}H_{ij}^T H_{ij} - \hat{\mathcal{D}}_{ij} - \frac{z_{ij}}{2} + g_{i(j-1)}z_{i(j-1)}\right) \tag{46}$$

$$\dot{\hat{\theta}}_{ij} = r_{ij}\left(\frac{1}{2\gamma_{ij}^2}H_{ij}^T H_{ij}z_{ij}^2 - \sigma_{ij}\hat{\theta}_{ij}\right) \tag{47}$$

Substituting (43)-(47) in to (42) results in

$$\begin{aligned}\dot{V}_{ij} \le& -k_{ij}z_{ij}^2 + \underbrace{\sigma_{ij}\tilde{\theta}_{ij}\hat{\theta}_{ij}}_{\text{III}} + g_{ij}z_{ij}z_{i(j+1)}\\ &+ \underbrace{z_{ij}\tilde{\mathcal{D}}_{ij} + \tilde{\mathcal{D}}_{ij}\dot{\mathcal{D}}_{ij} + c_{ij}\tilde{\mathcal{D}}_{ij}\tilde{\theta}_{ij}H_{ij}^T H_{ij}}_{\text{IV}} - c_{ij}\tilde{\mathcal{D}}_{ij}^2\\ &+ \sum_{k=1}^{2}\epsilon'_{ijk} + h_{ij}(t) + \frac{\gamma_{ij}^2}{2} - \frac{z_{ij}^2}{2} - g_{i(j-1)}z_{i(j-1)}z_{ij}\end{aligned} \tag{48}$$

Similar to (31) and (32), the following inequalities hold

$$\text{(III)}: \sigma_{ij}\tilde{\theta}_{ij}\hat{\theta}_{ij} \le -\frac{\sigma_{ij}\tilde{\theta}_{ij}^2}{2} + \frac{\sigma_{ij}\theta_{ij}^2}{2} \tag{49}$$

$$\begin{aligned}\text{(IV)}: z_{ij}\tilde{\mathcal{D}}_{ij} &\le \frac{z_{ij}^2}{2} + \frac{\tilde{\mathcal{D}}_{ij}^2}{2}\\ \tilde{\mathcal{D}}_{ij}\dot{\mathcal{D}}_{ij} &\le \frac{\tilde{\mathcal{D}}_{ij}^2}{2} + \frac{\rho_{ij}^2}{2}\\ c_{ij}\tilde{\mathcal{D}}_{ij}\tilde{\theta}_{ij}H_{ij}^T H_{ij} &\le \frac{\tilde{\mathcal{D}}_{ij}^2}{2} + \frac{\mu_{ij}^2 c_{ij}^2\tilde{\theta}_{ij}^2}{2}\end{aligned} \tag{50}$$

Based on (49) and (50), Eq. (48) satisfies the following inequality

$$\begin{aligned}\dot{V}_{ij} \le& -k_{ij}z_{ij}^2 - \frac{\sigma_{ij} - \mu_{ij}^2 c_{ij}^2}{2}\tilde{\theta}_{ij}^2 - (c_{ij} - 1.5)\tilde{\mathcal{D}}_{ij}^2\\ &+ g_{ij}z_{ij}z_{i(j+1)} + \sum_{k=1}^{2}\epsilon'_{ijk} + h_{ij}(t) + \frac{\rho_{ij}^2}{2} + \frac{\gamma_{ij}^2}{2}\\ &+ \frac{\sigma_{ij}\theta_{ij}^2}{2} - g_{i(j-1)}z_{i(j-1)}z_{ij}\end{aligned} \tag{51}$$

Step n_i: The actual control law v_i is designed in n_i step. Based on the coordination transformation $z_{in_i} = x_{in_i} - \alpha_{i(n_i-1)}$, the derivative of $\dot{z}_{in_i}$ is

$$\begin{aligned}\dot{z}_{in_i} &= g_{in_i}u_i(v_i) + f_{in_i}(X) + \Delta_{in_i} + d_{ij} - \dot{\alpha}_{i(n_i-1)}\\ &= g_{in_i}c_i v_i + g_{in_i}\iota_i(v_i) + f_{in_i}(X) + \Delta_{in_i} + d_{ij} - \dot{\alpha}_{i(n_i-1)}\end{aligned} \tag{52}$$

where $\dot{\alpha}_{i(n_i-1)}$ is defined in (37) with $j = n_i$. Choose a Lyapnov function as

$$V_{in_i} = \frac{1}{2}z_{in_i}^2 + \frac{c_i}{2r_{in_i}}\tilde{\theta}_{in_i}^2 + \frac{1}{2}\tilde{\mathcal{D}}_{in_i}^2 \tag{53}$$

Then, the time derivative of V_{in_i} is

$$\begin{aligned}\dot{V}_{in_i} \le& z_{in_i}\Big(g_{in_i}c_i v_i + g_{in_i}\iota_i(v_i) - \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}g_{ik}x_{i,(k+1)}\\ &+ f_{in_i}(X) - \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}f_{ik}(X) - \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}d_{ik}\\ &+ \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial\hat{\theta}_{ik}}\dot{\hat{\theta}}_{ik} + \frac{\partial\alpha_{i(n_i-1)}}{\partial r_i}\dot{r}_i\Big) + \left|z_{in_i}\bar{\Delta}_{in_i}\right| + z_{in_i}d_{in_i}\\ &- \frac{c_i}{r_{in_i}}\tilde{\theta}_{in_i}\dot{\hat{\theta}}_{in_i} + \tilde{\mathcal{D}}_{in_i}\dot{\tilde{\mathcal{D}}}_{in_i}\end{aligned} \tag{54}$$

where $\bar{\Delta}_{in_i} = \Delta_{in_i} - \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}\Delta_{ik}$.

Then, following the process from (39) to (42) outlined at step j, we can obtain the results below.

$$\begin{aligned}\left|z_{in_i}\bar{\Delta}_{in_i}\right| \le& z_{in_i}\hat{\varphi}_{in_i1}\left(\bar{x}_{in_i}\right) + z_{in_in_i}\hat{\varphi}_{in_i2}\left(\bar{x}_{in_i}, r_i\right)\\ &+ \frac{z_{in_i}^2}{4}\left[1 + \sum_{k=1}^{n_i-1}\left(\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}\right)^2\right]\\ &+ \epsilon'_{in_i1} + \epsilon'_{in_i2} + h_{in_i}(t)\end{aligned} \tag{55}$$

where $\hat{\varphi}_{in_i1}\left(\bar{x}_{in_i}\right), \hat{\varphi}_{in_i2}\left(\bar{x}_{in_i}, r_i\right), h_{in_i}(t)$ have been specified in (40) and (41) with $j = n_i$. Substituting (55) into (54) yields

$$\begin{aligned}\dot{V}_{in_i} \le& z_{in_i}\Big(g_{in_i}(\alpha_{ij} + z_{i(j+1)}) + \hat{f}_{in_i}(Z_{in_i})\Big) + z_{in_i}d_{in_i}\\ &- \frac{1}{r_{in_i}}\tilde{\theta}_{in_i}\dot{\hat{\theta}}_{in_i} + \tilde{\mathcal{D}}_{in_i}\dot{\tilde{\mathcal{D}}}_{in_i} + \sum_{k=1}^{2}\epsilon'_{in_ik} + h_{in_i}(t)\end{aligned} \tag{56}$$

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where

$$\begin{aligned}\hat{f}_{in_i}(Z_{in_i}) =& f_{in_i}(X) - \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}g_{ik}x_{i(k+1)} - \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}f_{ik}(X)\\ &- \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}d_{ik} + \sum_{k=1}^{n_i-1}\frac{\partial\alpha_{i(n_i-1)}}{\partial\hat{\theta}_{ik}}\dot{\hat{\theta}}_{ik} + \frac{\partial\alpha_{i(j-1)}}{\partial r_i}\dot{r}_i\\ &+ \hat{\varphi}_{in_i1}(\bar{x}_{in_i}) + \hat{\varphi}_{in_i2}(\bar{x}_{in_i}, r_i) + \frac{z_{in_i}^2}{4}\left[1 + \sum_{k=1}^{n_i-1}\left(\frac{\partial\alpha_{i(n_i-1)}}{\partial x_{ik}}\right)^2\right]\\ Z_{in_i} =& [\bar{x}_{in_i}, \bar{\hat{\theta}}_{i(n_i-1)}, r]^T \in \Omega_{Z_{in_i}} \subset R^{2n_i}. \end{aligned} \tag{57}$$

By employing an RBFNN $W_{in_i}^T H_{in_i}(Z_{in_i})$ to approximate $\hat{f}_{in_i}(Z_{in_i})$, the following equation can be obtained

$$\hat{f}_{in_i}(Z_{in_i}) = W_{in_i}^T H_{in_i}(Z_{in_i}) + \xi_{in_i} \tag{58}$$

Define the following disturbance observer:

$$\begin{cases}\mathcal{D}_{in_i} = d_{in_i} + \xi_{in_i}\\ \hat{\mathcal{D}}_{in_i} = \hat{\Xi}_{in_i} + c_{in_i}z_{in_i}\\ \dot{\hat{\Xi}}_{in_i} = -c_{in_i}\left(\hat{\theta}_{in_i}H_{in_i}^T H_{in_i} + g_{in_i}u_i + \hat{\mathcal{D}}_{in_i}\right)\\ \dot{\tilde{\mathcal{D}}}_{in_i} = \dot{\mathcal{D}}_{in_i} - c_{in_i}\left(\tilde{\mathcal{D}}_{in_i} - \tilde{\theta}_{in_i}H_{in_i}^T H_{in_i}\right)\end{cases} \tag{59}$$

where $\tilde{\mathcal{D}}_{in_i} = \mathcal{D}_{in_i} - \hat{\mathcal{D}}_{in_i}$ is the estimate error of disturbance, $\dot{\mathcal{D}}_{in_i} \le \rho_{in_i}, \rho_{in_i} > 0$ is an unknown constant, Ξ_{in_i} is the auxiliary variable. Then, the virtual control input and adaption laws shall be designed as

$$\begin{aligned}v_i =& g_{in_i}^{-1}\Big(-k_{in_i}z_{in_i} - \frac{z_{in_i}}{2\gamma_{in_i}^2}\hat{\theta}_{in_i}H_{in_i}^T H_{in_i} - c_i^{-1}\hat{\mathcal{D}}_{in_i}\\ &- c_i^{-1}z_{in_i} + c_i^{-1}g_{i(n_i-1)}z_{i(n_i-1)}\Big)\end{aligned} \tag{60}$$

$$\dot{\hat{\theta}}_{in_i} = r_{in_i}\left(\frac{1}{2\gamma_{in_i}^2}H_{in_i}^T H_{in_i}z_{in_i}^2 - \sigma_{in_i}\hat{\theta}_{in_i}\right) \tag{61}$$

Substituting (57)-(61) in to (56) results in

$$\begin{aligned}\dot{V}_{in_i} \le& -k_{in_i}c_iz_{in_i}^2 + \underbrace{c_i\sigma_{in_i}\tilde{\theta}_{in_i}\hat{\theta}_{in_i}}_{\text{V}}\\ &+ \underbrace{z_{in_i}\tilde{\mathcal{D}}_{in_i} + \tilde{\mathcal{D}}_{in_i}\dot{\mathcal{D}}_{in_i} + c_{in_i}\tilde{\mathcal{D}}_{in_i}\tilde{\theta}_{in_i}H_{in_i}^T H_{in_i}}_{\text{VI}} - c_{in_i}\tilde{\mathcal{D}}_{in_i}^2\\ &+ \sum_{k=1}^{2}\epsilon'_{in_ik} + h_{in_i}(t) + \frac{\gamma_{in_i}^2}{2} - z_{in_i}^2 + \underbrace{z_{in_i}g_{in_i}\iota_i(v_i)}_{\text{VII}}\\ &- g_{i(n_i-1)}z_{i(n_i-1)}z_{in_i}\end{aligned} \tag{62}$$

Similar to (49) and (50), the following inequalities hold

$$\text{(V)}: c_i\sigma_{in_i}\tilde{\theta}_{in_i}\hat{\theta}_{in_i} \le -\frac{c_i\sigma_{in_i}\tilde{\theta}_{in_i}^2}{2} + \frac{c_i\sigma_{in_i}\theta_{in_i}^2}{2} \tag{63}$$

$$\begin{aligned}\text{(VI)}: z_{in_i}\tilde{\mathcal{D}}_{in_i} \le& \frac{z_{in_i}^2}{2} + \frac{\tilde{\mathcal{D}}_{in_i}^2}{2}\\ \tilde{\mathcal{D}}_{in_i}\dot{\mathcal{D}}_{in_i} \le& \frac{\tilde{\mathcal{D}}_{in_i}^2}{2} + \frac{\rho_{in_i}^2}{2}\\ c_{in_i}\tilde{\mathcal{D}}_{in_i}\tilde{\theta}_{in_i}H_{in_i}^T H_{in_i} \le& \frac{\tilde{\mathcal{D}}_{in_i}^2}{2} + \frac{\mu_{in_i}^2c_{in_i}^2\tilde{\theta}_{in_i}^2}{2}\end{aligned} \tag{64}$$

$$\text{(VII)}: z_{in_i}g_{in_i}\iota_i(v_i) \le \frac{z_{in_i}^2}{2} + \frac{\bar{g}_{in_i}\bar{\iota}_i(v_i)^2}{2} \tag{65}$$

Based on (63)-(65), one has

$$\begin{aligned}\dot{V}_{in_i} \le& -k_{in_i}c_iz_{in_i}^2 - \frac{c_i\sigma_{in_i} - \mu_{in_i}^2c_{in_i}^2}{2}\tilde{\theta}_{in_i}^2 - (c_{in_i} - 1.5)\tilde{\mathcal{D}}_{in_i}^2\\ &+ \sum_{k=1}^{2}\epsilon'_{in_ik} + h_{in_i}(t) + \frac{\rho_{in_i}^2}{2} + \frac{\gamma_{in_i}^2}{2}\\ &+ \frac{c_i\sigma_{in_i}\theta_{in_i}^2}{2} - g_{i(n_i-1)}z_{i(n_i-1)}z_{in_i} + \frac{\bar{g}_{in_i}\bar{\iota}_i(v_i)^2}{2}\end{aligned} \tag{66}$$

At current stage, a Lyapunov function is selected for the overall systems as

$$\begin{aligned}V =& \sum_{i=1}^{N}\sum_{j=1}^{n_i}V_{ij}\\ =& \sum_{i=1}^{N}\left(\sum_{j=1}^{n_i}\frac{z_{ij}^2}{2} + \sum_{j=1}^{n_i-1}\frac{\tilde{\theta}_{ij}^2}{2r_{ij}} + \sum_{j=1}^{n_i}\frac{\tilde{\mathcal{D}}_{ij}^2}{2} + \frac{r_i}{\lambda_{i0}} + \frac{c_i\tilde{\theta}_{in_i}^2}{2r_{in_i}}\right)\end{aligned} \tag{67}$$

By the inequalities (33), (51) and (66), we get

$$\begin{aligned}\dot{V} \le& \sum_{i=1}^{N}\Big\{-k_{i1}z_{i1}^2 - \frac{\sigma_{i1} - \mu_{i1}^2c_{i1}^2}{2}\tilde{\theta}_{i1}^2 + g_{i1}z_{i1}z_{i2}\\ &- (c_{i1} - 1.5)\tilde{\mathcal{D}}_{i1}^2 + \left(1 - 2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}}\\ &+ \frac{d_{i0}}{\lambda_{i0}} - \frac{\bar{c}_i}{\lambda_{i0}}r_i + \sum_{k=1}^{2}\epsilon'_{i1k} + h_{i1}(t) + \frac{\rho_{i1}^2}{2} + \frac{\gamma_{i1}^2}{2} + \frac{\sigma_{i1}\theta_{i1}^2}{2}\Big\}\\ &+ \sum_{i=1}^{N}\sum_{j=2}^{n_i-1}\Big\{-k_{ij}z_{ij}^2 - \frac{\sigma_{ij} - \mu_{ij}^2c_{ij}^2}{2}\tilde{\theta}_{ij}^2 - (c_{ij} - 1.5)\tilde{\mathcal{D}}_{ij}^2\\ &+ g_{ij}z_{ij}z_{i(j+1)} + \sum_{k=1}^{2}\epsilon'_{ijk} + h_{ij}(t) + \frac{\rho_{ij}^2}{2} + \frac{\gamma_{ij}^2}{2}\\ &+ \frac{\sigma_{ij}\theta_{ij}^2}{2} - g_{i(j-1)}z_{i(j-1)}z_{ij}\Big\} + \sum_{i=1}^{N}\Big\{-k_{in_i}c_iz_{in_i}^2\\ &- \frac{c_i\sigma_{in_i} - \mu_{in_i}^2c_{in_i}^2}{2}\tilde{\theta}_{in_i}^2 - (c_{in_i} - 1.5)\tilde{\mathcal{D}}_{in_i}^2\\ &+ \sum_{k=1}^{2}\epsilon'_{in_ik} + h_{in_i}(t) + \frac{\rho_{in_i}^2}{2} + \frac{\gamma_{in_i}^2}{2}\\ &+ \frac{\sigma_{in_i}\theta_{in_i}^2}{2} - g_{i(n_i-1)}z_{i(n_i-1)}z_{in_i} + \frac{\bar{g}_{in_i}\bar{\iota}_i(v_i)^2}{2}\Big\}\end{aligned} \tag{68}$$

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Next, by rearranging the sequence, one has

$$\dot{V} \le \sum_{i=1}^{N}\Bigg\{\sum_{j=1}^{n_i-1} -k_{ij}z_{ij}^2 - k_{in_i}c_i z_{in_i}^2 - \frac{\bar{c}_i}{\lambda_{i0}}r_i - \sum_{j=1}^{n_i-1}\frac{\sigma_{ij}-\mu_{ij}^2 c_{ij}^2}{2}\tilde{\theta}_{ij}^2 - \frac{c_i\sigma_{in_i}-\mu_{in_i}^2 c_{in_i}^2}{2}\tilde{\theta}_{in_i}^2 - \sum_{j=1}^{n_i}(c_{ij}-1.5)\tilde{\mathcal{D}}_{ij}^2 + \sum_{j=1}^{n_i} S_{ij} + \left(1-2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}} + \frac{d_{i0}}{\lambda_{i0}} + \frac{\bar{g}_{in_i}\bar{l}_i(v_i)^2}{2}\Bigg\} \tag{69}$$

Theorem 1. *Consider the nonlinear system (1) with the controllers (60) and adaptive update laws (29), (47), (61). The nonlinear disturbance observers are proposed as (25), (45), (59). For $1 \le i \le N$, $1 \le j \le n_i$, assume that all the unknown nonlinear functions $\hat{f}_{i,j}(Z)$ are approximated by neural networks in the sense that the approximation error $\xi_{i,j}$ is bounded. Then, for bounded initial conditions, under the developed control laws (60), all signals of the closed-loop system are bounded and the target signal can be tracked within a small bounded error.*

Proof. Let

$$\rho = \min\Big\{2k_{ij}, \bar{c}_i, r_{ij}(\sigma_{ij}-\mu_{ij}^2c_{ij}^2), 2(c_{ij}-1.5), 2k_{in_i}c_i, r_{in_i}(c_i\sigma_{in_i}-\mu_{in_i}^2c_{in_i}^2), 1\le i\le N, 1\le j\le n_i-1\Big\} \tag{70}$$

and

$$\varsigma_0 = \sum_{i=1}^{N}\sum_{j=1}^{n_i} S_{ij} + \frac{d_{i0}}{\lambda_{i0}} + \frac{\bar{g}_{in_i}\bar{l}_i(v_i)^2}{2} \tag{71}$$

such that (69) is rewritten as

$$\dot{V} \le -\rho V + \varsigma_0 + \sum_{i=1}^{N}\left(1-2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}}, t\ge 0. \tag{72}$$

It is noted that the last term $\left(1-2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}}$ in (72) may be positive or negative which depends on the size of z_{i1}. Therefore, the following two cases should be considered based (72).

Case 1: $z_{i1} \in \Omega_{z_{i1}} = \{z_{i1}||z_{i1}| < 0.8814\tau_i\}$ for any positive constant τ_i. By using Assumption 4 and Lemma 1, $\mu_0(x_{i1}^2)$ is a non-negative smooth function, $\left(1-2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}}$ is bounded, there exists a positive constant $\mu_{i0} > 0$, such that $\left(1-2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}} \le \mu_{i0}$. From (72), we have

$$\dot{V} \le -\rho V + b_0 \tag{73}$$

where $b_0 = \varsigma_0 + \sum_{i=1}^{N}\mu_{i0}$. Furthermore, (73) satisfies

$$V(t) \le \left(V(0)-\frac{b_0}{\rho}\right)e^{-\rho t} + \frac{b_0}{\rho}, t\ge 0. \tag{74}$$

Case 2: $z_{i1} \notin \Omega_{z_{i1}}$. By using Lemma 3 and the fact that $\frac{x_{i1}^2\mu_{i0}(x_{i1}^2)}{\lambda_{i0}} \ge 0$, we can obtain

$$\left(1-2\tanh^2\left(\frac{z_{i1}}{\tau_i}\right)\right)\frac{x_{i1}^2\mu_0(x_{i1}^2)}{\lambda_{i0}} \le 0 \tag{75}$$

Therefore, (72) is simplified as

$$\dot{V} \le -\rho V + \varsigma_0 \tag{76}$$

Furthermore, (76) satisfies

$$V(t) \le \left(V(0)-\frac{\varsigma_0}{\rho}\right)e^{-\rho t} + \frac{\varsigma_0}{\rho}, t\ge 0. \tag{77}$$

Consider (74) and (77), we obtain

$$V(t) \le \left(V(0)-\frac{b_0}{\rho}\right)e^{-\rho t} + \frac{b_0}{\rho}, t\ge 0. \tag{78}$$

1) From (78), according to the boundedness theorem (e.g.,[57]), we have that all z_i and $\hat{\theta}_i (i = 1,\dots,n)$ are uniformly ultimately bounded. Since $z_{i1} = x_{i1} - y_{ri}$ and y_{ri} are bounded, we have that x_{i1} is bounded. From $z_{ij} = x_{ij} - \alpha_{i(j-1)}$, $j = 1,\dots,n_i$, and the virtual control laws (28), (46), we have that x_{ij}, $j = 2,\dots,n_i$ remain bounded. Using (60) we conclude that control v_i is also bounded. Thus, all the signals in the closed-loop system remain bounded.

2) From (78), we have

$$\sum_{i=1}^{N}\frac{1}{2}z_i^2 \le \left(V(0)-\frac{b_0}{\rho}\right)e^{-\rho t} + \frac{b_0}{\rho} < V(0)e^{-\rho t} + \frac{b_0}{\rho} \tag{79}$$

that is

$$\sum_{i=1}^{N} z_i^2 < 2V(0)e^{-\rho t} + \frac{2b_0}{\rho} \tag{80}$$

In particular, we have

$$\lim_{t\to\infty}|z_{i1}| = \lim_{t\to\infty}|y_i(t)-y_{ri}(t)| \le \sqrt{\frac{2b_0}{\rho}} \tag{81}$$

which implies that the system output is ensured to converge to a small bounded error. The proof is completed here. □

Table 1
Initial values for example 1

$x_{1,1}(0) = 1$	$x_{1,2}(0) = -1$	$x_{2,1}(0) = 0.5$	$x_{2,2}(0) = -0.5$
$\hat{\theta}_{1,1}(0) = 0.01$	$\hat{\theta}_{1,2}(0) = 0.1$	$\hat{\theta}_{2,1}(0) = 0.01$	$\hat{\theta}_{2,2}(0) = 0.1$
$\alpha_{1,1} = 0.01$	$\alpha_{1,2} = 0.02$	$\alpha_{2,1} = 0.01$	$\alpha_{2,2} = 0.02$

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Table 2
Controller parameters for example 1

$k_{11} = 22$	$k_{12} = 40$	$k_{21} = 22$	$k_{22} = 40$	$c_{11} = 5$
$c_{12} = 5$	$c_{21} = 5$	$c_{22} = 5$	$\gamma_{11} = 8$	$\gamma_{12} = 10$
$\gamma_{21} = 8$	$\gamma_{22} = 10$	$\sigma_{11} = 0.02$	$\sigma_{12} = 0.03$	$\sigma_{21} = 0.02$
$\sigma_{22} = 0.03$	$r_{11} = 20$	$r_{12} = 50$	$r_{21} = 20$	$r_{22} = 45$

Table 3
Initial values for example 2

$x_{1,1}(0) = 1$	$x_{1,2}(0) = -1$	$x_{2,1}(0) = 0.5$	$x_{2,2}(0) = -0.5$
$x_{3,1}(0) = 0.5$	$x_{3,2}(0) = 1$	$\hat{\theta}_{1,1}(0) = 0.01$	$\hat{\theta}_{1,2}(0) = 0.1$
$\hat{\theta}_{2,1}(0) = 0.01$	$\hat{\theta}_{2,2}(0) = 0.1$	$\hat{\theta}_{3,1}(0) = 0.01$	$\hat{\theta}_{3,2}(0) = 0.1$
$\alpha_{1,1} = 0.01$	$\alpha_{1,2} = 0.02$	$\alpha_{2,1} = 0.01$	$\alpha_{2,2} = 0.02$
$\alpha_{3,1} = 0.01$	$\alpha_{3,2} = 0.02$		

4. Simulation studies

In this section, two simulation examples are presented to show the effectiveness of the developed method.

Example 1: Consider the following MIMO systems composed of three subsystems with input dead-zone:

$$\begin{cases} \dot{z}_1 = -z_1 + x_{11}^2 + 0.5 \\ \dot{x}_{11} = f_{11}(\cdot) + g_{11}(\cdot)x_{12} + d_{11}(t) \\ \dot{x}_{12} = f_{12}(\cdot) + g_{12}(\cdot)u_1(v_1) + d_{12}(t) \\ \dot{z}_2 = -z_2 + x_{21}^2 + 0.5 \\ \dot{x}_{21} = f_{21}(\cdot) + g_{21}(\cdot)x_{22} + d_{21}(t) \\ \dot{x}_{22} = f_{22}(\cdot) + g_{22}(\cdot)u_2(v_2) + d_{22}(t) \\ y_1 = x_{11}, y_2 = x_{21} \end{cases} \tag{82}$$

where the external disturbances are chosen as

$$D_1(t) = \begin{bmatrix} 0.2\sin(t) + sin(1.5t) \\ 0.1\sin(t) \end{bmatrix}$$
$$D_2(t) = \begin{bmatrix} 0.3\sin(t) + \sin(1.2t) \\ 0.1\sin(t) \end{bmatrix} \tag{83}$$

where $f_{i1}(\cdot) = 0, g_{i1}(\cdot) = 1, i = 1,2, g_{12}(\cdot) = 1, f_{12}(\cdot) = 20\sin x_{11} - 8\sin x_{11}\cos x_{11} + 8\sin x_{21}\cos_{21}, g_{22}(\cdot) = 0.7, f_{22}(\cdot) = 20\sin x_{21} - 10.6\sin x_{21}\cos x_{21} + 5.3\sin x_{11}\cos x_{11}, u_i, i = 1,2$, are the actual inputs of the MIMO systems. The outputs of the dead- zone actuators $u_i, i = 1,2$, are described as

$$u_i = G(v_i(t)) = \begin{cases} (1 - 0.2\sin(v_i)(v_i - 1)), v_i(t) \geq 1 \\ 0, -0.5 \leq v_i(t) \leq 1 \\ (0.8 - 0.1\cos(v_i))(v_i + 0.5), v_i(t) \leq -0.5 \end{cases} \tag{84}$$

The desired output is assumed to be

$$\begin{cases} y_{1d} = 0.5\sin 2\pi t \\ y_{2d} = 0.8\sin \pi t \end{cases} \tag{85}$$

The initial values and the design parameters are shown in Table 1 and Table 2, respectively. Neural networks $W_{i1}^T H_{i1}(Z_{i1})$ contains 11 nodes (i.e., l_{i1}=11), with centers $C_l(l = 1,\ldots,l_{i1})$ evenly spaced in $[-10, 10] \times [-10, 10]$, and widths $b_l = 1(l = 1,\ldots,l_{i1})$. Neural networks $W_{i2}^T H_{i2}(Z_{i2})$ contains 21 nodes (i.e., l_{i2}=21), with centers $C_l(l = 1,\ldots,l_{i2})$ evenly spaced in [-20,20]×[-20,20]×[-20,20]×[-20,20] and widths $b_l = 1.5(l = 1,\ldots,l_{i2})$.

The simulation figures are listed in Figs. 1-6. Fig. 1 demonstrate that x_{11}, x_{21} tracks their references effectively and tracking errors are shown in Fig. 2. The control inputs u are shown in Fig. 3. The disturbance estimate result is in Figs. 4-5. Figs. 4-5 indicate that the proposed disturbance observer can estimate external disturbance within a small bounded error. The adaptive parameters θ_i $i = 1,2$ are shown in Fig. 6, respectively. From the above-mentioned simulation figures, we can see that the proposed approach in this paper provides good transient performance and the tracking error is as good as desired, while all the signals of the closed-loop system are bounded.

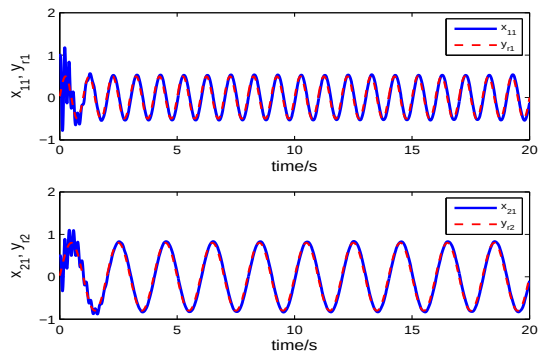

Fig. 1. Trajectories of the outputs x_{i1}, the reference signals $y_{ri}, i = 1,2$.

Example 2: Consider the model of three-axis ship-mounted satellite tracking antenna systems [58]:

$$\begin{cases} 5.7\ddot{q}_1 + 16.2\dot{q}_1 - 5.7\dot{q}_2\dot{q}_3\cos q_2 + 7.2\dot{q}_3\sin q_2 = u_1 \\ 38.6\ddot{q}_2 + 33.1\dot{q}_2 + 5.2\dot{q}_1\dot{q}_3\cos q_2 + 25.4\dot{q}_3^2\sin q_2\cos q_2 = u_2 \\ 54.7\ddot{q}_3 + 78.3\dot{q}_3 - 14.1\dot{q}_2\dot{q}_3\sin q_2\cos q_2 - 1.4\dot{q}_1\dot{q}_2\cos q_2 = u_3 \end{cases} \tag{86}$$

where $q = [q_1, q_2, q_3]^T$ are pitch, roll, and yaw angle respectively, u denotes the control input. In addition, there always

Table 4
Controller parameters for example 2

$k_{11} = 17$	$k_{12} = 50$	$k_{21} = 25$	$k_{22} = 50$	$k_{31} = 26$
$k_{32} = 50$	$c_{11} = 3$	$c_{12} = 3$	$c_{21} = 0.05$	$c_{22} = 0.05$
$c_{31} = 0.01$	$c_{32} = 0.01$	$\gamma_{11} = 1$	$\gamma_{12} = 1$	$\gamma_{21} = 5$
$\gamma_{22} = 5$	$\gamma_{31} = 4$	$\gamma_{32} = 4$	$\sigma_{11} = 0.8$	$\sigma_{12} = 0.8$
$\sigma_{21} = 0.08$	$\sigma_{22} = 0.08$	$\sigma_{31} = 0.5$	$\sigma_{32} = 0.5$	$r_{11} = 8$
$r_{12} = 8$	$r_{21} = 8$	$r_{22} = 10$	$r_{31} = 10$	$r_{32} = 10$
$\rho_{11} = 20$	$\rho_{12} = 50$	$\rho_{21} = 20$	$\rho_{22} = 48$	$\rho_{31} = 18$
$\rho_{32} = 52$				

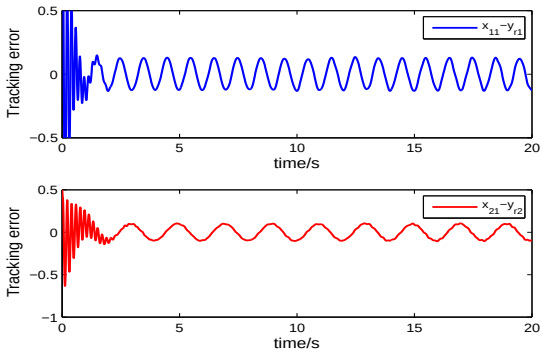

Fig. 2. The tracking errors $x_{i1} - y_{ri}, i = 1, 2$.

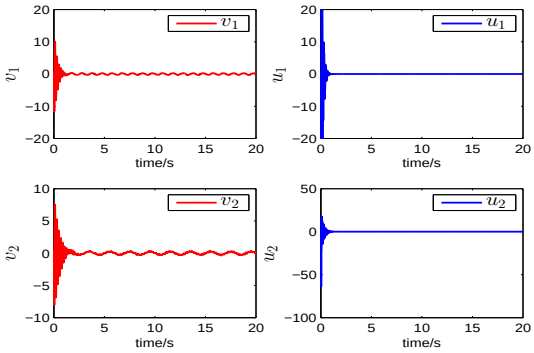

Fig. 3. Trajectories of the output u_i and input $v_i, i = 1, 2$ of dead-zone.

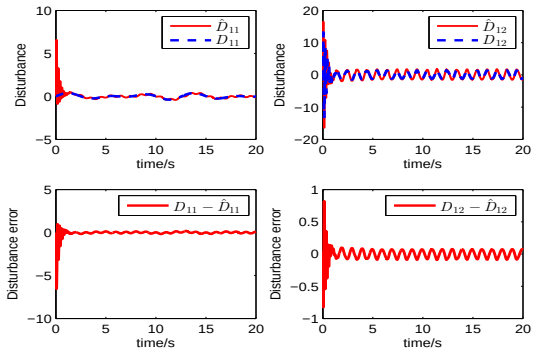

Fig. 4. Disturbance estimate for D_1 under the proposed control strategy.

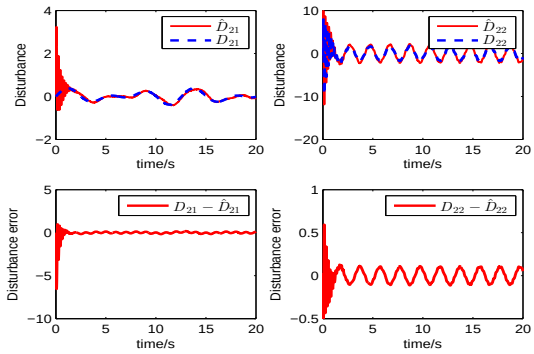

Fig. 5. Disturbance estimate for D_2 under the proposed control strategy.

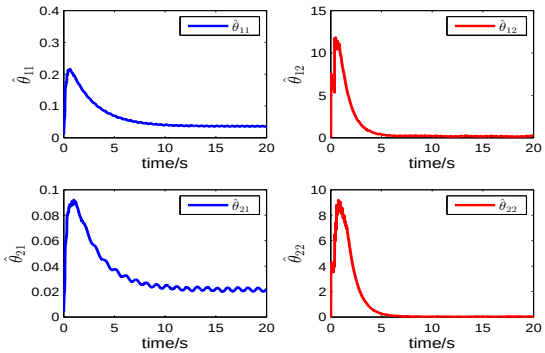

Fig. 6. Trajectories of adaptive laws $\hat{\theta}_1, \hat{\theta}_2$.

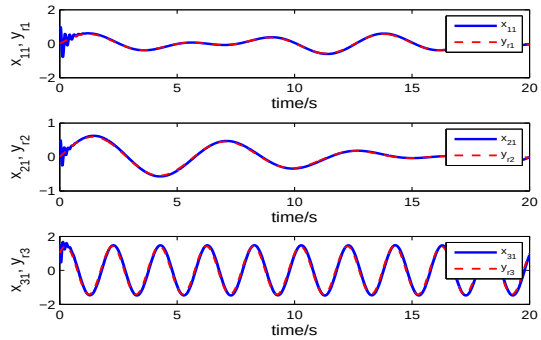

Fig. 7. Trajectories of the outputs x_{i1}, the reference signals $y_{ri}, i = 1, 2, 3$ for Example 2.

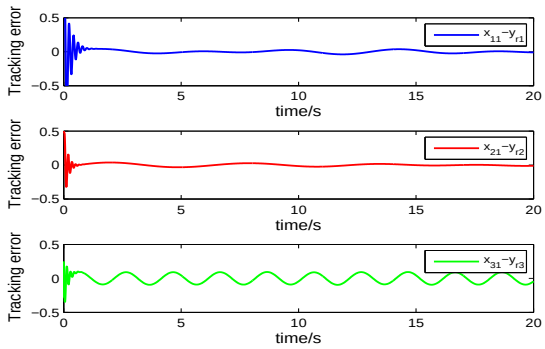

Fig. 8. The tracking errors $x_{i1} - y_{ri}, i = 1, 2, 3$ for Example 2.

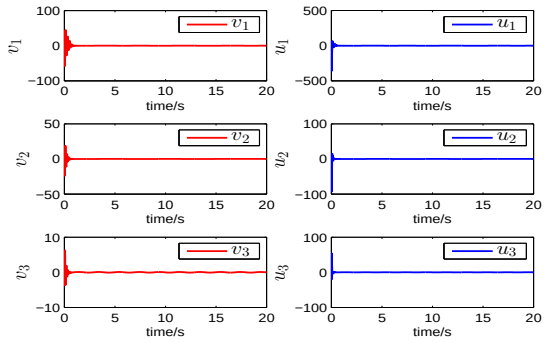

Fig. 9. Trajectories of the output u_i and input $v_i, i = 1, 2, 3$ of dead-zone for Example 2.

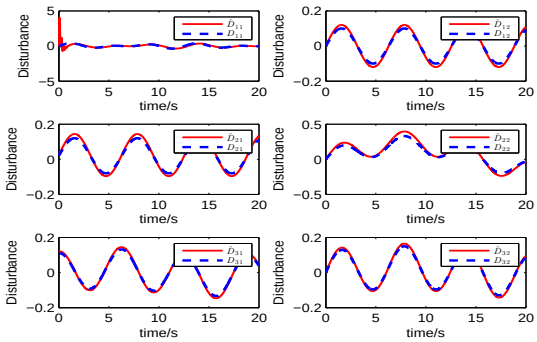

Fig. 10. Disturbance estimate $\hat{D}_1, \hat{D}_2, \hat{D}_3$ under the proposed control strategy for Example 2.

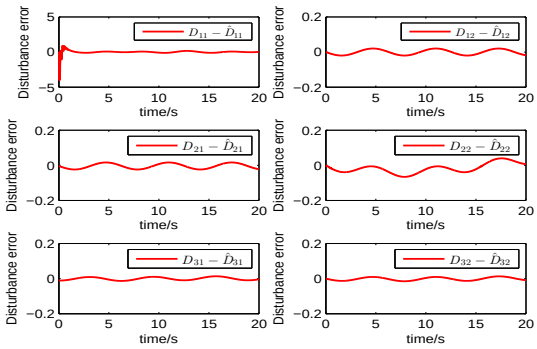

Fig. 11. Disturbance estimate error under the proposed control strategy for Example 2.

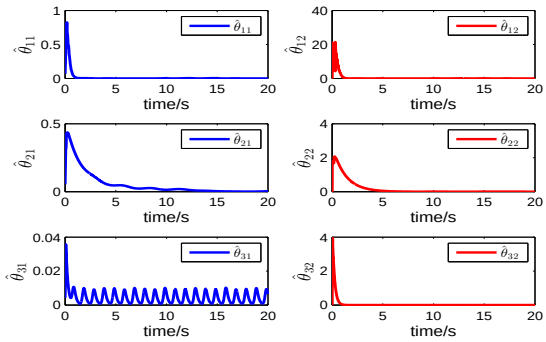

Fig. 12. Trajectories of adaptive laws $\hat{\theta}_1, \hat{\theta}_2, \hat{\theta}_3$ for Example 2.

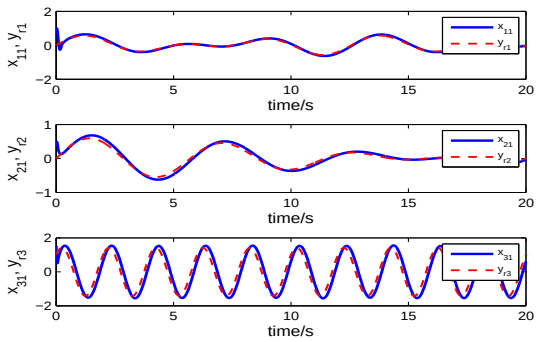

Fig. 13. Trajectory of $x_{i1}, y_{ri}, i = 1, 2, 3$ using controller u^* in [44] for Example 2.

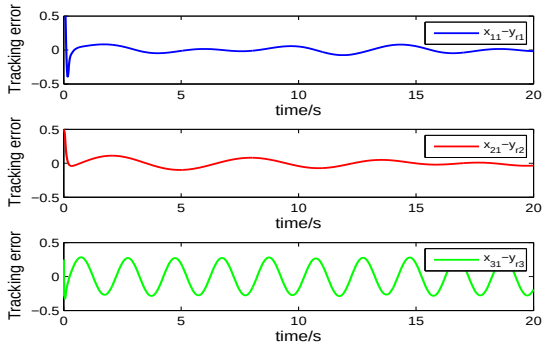

Fig. 14. The tracking errors $x_{i1} - y_{ri}, i = 1, 2, 3$ using controller u^* in [44] for Example 2.

exists external time-varying disturbances and system uncertainties. The unknown input dead-zone in actuator is another problem which may be encountered, all of these factors bring out model uncertainties and difficulties in control of ship-mounted satellite tracking system. In order to evaluate the proposed method, we must first change the ship-mounted satellite model (86) into the shape of (1), that is

$$\begin{cases} \dot{z}_1 = -z_1 + 0.5x_{11}^2 + 0.5 \\ \dot{x}_{11} = x_{12} \\ \dot{x}_{12} = x_{22}x_{32}\cos x_{21} - 2.84x_{12} - 1.26x_{32}\sin x_{21} + 0.18u_1 \\ \dot{z}_2 = -z_2 + 0.5x_{21}^2 + 0.5 \\ \dot{x}_{21} = x_{22} \\ \dot{x}_{22} = -0.86x_{22} - 0.13x_{12}x_{32}\cos x_{21} - 0.33x_{32}^2\sin 2x_{21} + 0.03u_2 \\ \dot{z}_3 = -z_3 + 0.5x_{31}^2 + 0.5 \\ \dot{x}_{31} = x_{32} \\ \dot{x}_{32} = 0.13x_{22}x_{32}\sin 2x_{21} + 0.03x_{12}x_{22}\cos x_{21} - 1.43x_{32} + 0.02u_3 \\ y_1 = x_{11}, y_2 = x_{21}, y_3 = x_{31} \end{cases} \tag{87}$$

where $x_1 = [q_1, \dot{q}_1]^T, x_2 = [q_2, \dot{q}_2]^T, x_3 = [q_3, \dot{q}_3]^T$ and the external disturbances are chosen as

$$D_1(t) = \begin{bmatrix} 0.2\sin(t) + \sin(1.5t) \\ 0.1\sin(t) \end{bmatrix}$$
$$D_2(t) = \begin{bmatrix} 0.1\sin(0.01t) + 0.2\cos(0.01t) \\ 0.13\sin(t) + 0.05\sin(0.2t) \end{bmatrix}$$
$$D_3(t) = \begin{bmatrix} 0.11\cos(t) + 0.21\sin(0.3t) \\ 0.12\sin(t) + 0.03\sin(0.2t) \end{bmatrix} \tag{88}$$

We choose the desired output

$$\begin{cases} y_{1d} = 0.3\sin(t) + \sin(1.5t) \\ y_{2d} = 0.2\sin(t) + \sin(1.2t) \\ y_{3d} = \sin\pi t + \cos\pi t \end{cases} \tag{89}$$

The initial values and the design parameters are shown in Table 3 and Table 4, respectively. The simulation figures are listed in Figs. 7-12. Fig. 7 demonstrates that x_{11}, x_{21}, x_{31} tracks their references effectively and tracking errors are shown in Fig.

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8. The control inputs u are shown in Fig. 9. The disturbance estimate result is in Fig. 10. Fig. 11 indicates that the proposed disturbance observer can estimate external disturbance within a small bounded error. The adaptive parameters $\theta_i, i = 1, 2, 3$ is shown in Fig. 12. In addition, the adaptive sliding model controller u^* developed in [44] is applied to further show the advantage of the control scheme proposed in this paper. The simulation results are shown in Fig. 13 and 14, from Fig. 13 it can be seen that the tracking performance is very poor and Fig. 14 indicates that our method can ensure the target signal be tracked within a small bounded error.

5. Conclusion

In this paper, disturbance observer-based adaptive neural tracking control is proposed for a class of multiple-input multiple-output systems in the presence of unmodeled dynamics, system uncertainties, time varying disturbance and input dead-zone. The theory analysis and simulation results indicate that the target signal can be converged to a small bounded error. In the future researches, we will present a control algorithm for system (1) with both state and input constraints, and pay attention to more complex systems such as stochastic time-delay nonlinear systems.

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